Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: PixelTrendAnalytics
b. Team leader name: Muthukumar Arunachalam
c. Problem Statement: Trace quality evidence into a reviewed inspection plan.

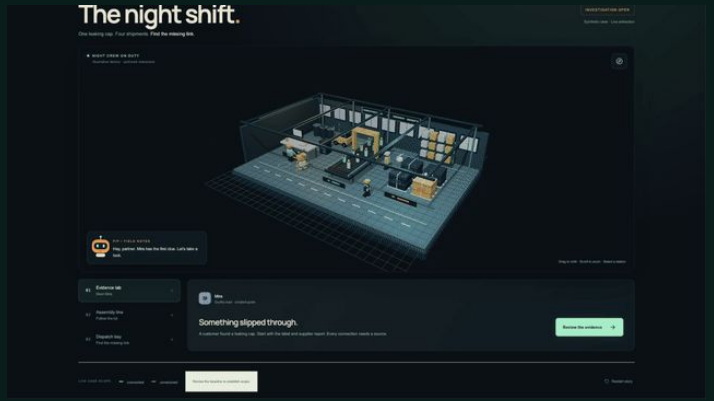
Brief about the idea

One leaking cap.
Possible causes.
One next check.

Pip investigates.

Operators test and review.

Gemini reassesses.



Synthetic case. Real Gemini. Human-reviewed decisions.

Opportunities

DIFFERENCE: Gemini revises hypotheses after operator results.
SOLUTION: Evidence, inspection missions and reviewed actions.
USP: A playable factory with an auditable agent journal.

An investigation that learns from results.

Pip proposes missions and revises evidence-backed hypotheses.

Operators approve checks, facts and the final handover.

List of features offered by the solution



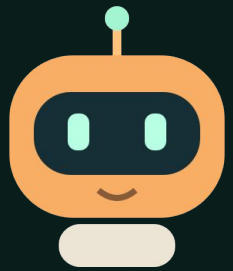
Mira’s evidence lab

Read evidence. Compare causes. Approve a demo inspection mission and record its result.



Sol’s assembly line

Review assembly records. Compare inspection scopes with explicit workload assumptions.



Pip’s investigation

Reassess results with Gemini. Ask operational questions. Review corrective-action drafts.

Process flow diagram or Use-case diagram

01 INVESTIGATE
Compare causes

Read evidence and procedures

Find gaps and contradictions

Propose inspection missions

→

02 INSPECT
Operator checkpoint

Approve a demo mission

Record a synthetic result

Gemini revises its findings

→

03 HAND OVER
Review next steps

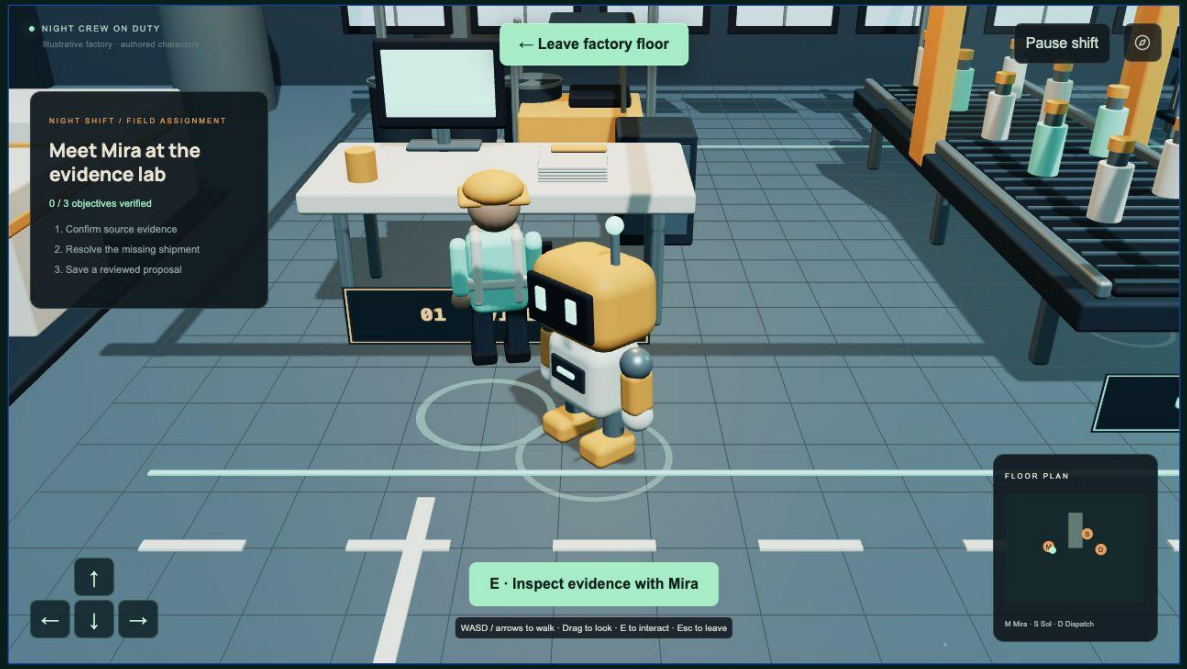
Compare inspection scopes

Approve action drafts

Export a reviewed snapshot

Operational chat clarifies the case using cited evidence.
Humans approve decisions. Demo results are explicitly synthetic.

Wireframes/Mock diagrams of the proposed solution (optional)



Walk as Pip. Approach a station. Press E to enter the real evidence-review workflow.

Architecture diagram of the proposed solution



Operations-only Gemini chat uses a separate scope gate and has no mutation tools.

Technologies to be used in the solution

Gemini 3.5 Flash-Lite via Vertex AI
Tool calling, cause comparison and grounded operational chat.

Cloud Run + Firestore
Bounded execution, mission results and persistent case journals.

React, TypeScript, Three.js + FastAPI
Playable factory, human approvals and deterministic scope.

Estimated implementation cost (optional)

Pilot estimate: pending

0 minimum / 2 maximum Cloud Run instances

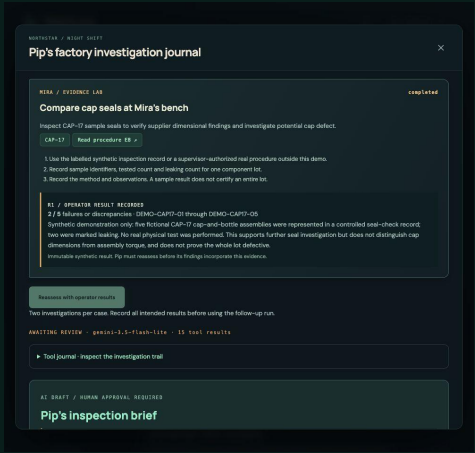
100 model attempts/day across cases

Per case: 12 investigator + 8 scans + 12 chat calls

Per run: 6 turns / 24 tools / 95 seconds

Optional estimate not yet measured. Cloud usage depends on tokens, requests and storage. These controls limit usage; they are not a monetary spending cap.

Snapshots of the prototype



Pip proposes.
Operators inspect.

Gemini reassesses.
Humans decide.

Cause comparisons, inspection missions and operational chat with real Gemini.

Prototype Performance report/Benchmarking

1,500

units proposed for inspection

Zero unresolved after review

ADAPTIVE QUALITY INVESTIGATION

Cause comparison and missions

22 backend tests passed

Human approval gates verified

Synthetic case and operator-entered demo results. No production accuracy claim.

Gemini recommendations cannot approve scope, release stock or operate equipment.

Additional Details/Future Development (if any)

Next shift: a real-team pilot

Can a quality team reach a defensible decision with less searching and fewer missed connections?

Measure evidence coverage, review time and decision errors.

Next: approved real procedures, controlled uploads and ERP integration.

Try BatchLens and inspect the source

- GitHub: https://github.com/mk-tdev/BatchLens
- Demo video: recording link pending
- Live: https://batchlens-hk3hxwi5na-as.a.run.app/

Walk the factory.
Complete a mission.
Challenge Pip.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

